**University of Parma**
Department of Mathematical, Physical and Computer Sciences
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# Quantum simulation of molecular spin systems on quantum hardware

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### Abstract

Molecular magnets are privileged platforms for studying collective quantum phenomena, yet their classical simulation quickly becomes intractable as the number of spins increases. Quantum computers, by mapping each spin-1/2 directly onto a qubit, offer a natural route around this bottleneck. This work presents the quantum simulation, on both ideal and realistically noisy simulators, of spin-1/2 systems coupled through a Heisenberg interaction with an anisotropic Dzyaloshinskii–Moriya term, starting from the simplest case (the dimer) up to a frustrated three-spin ring. The ground state is determined with the Variational Quantum Eigensolver, comparing a generic hardware-efficient ansatz with ansätze built on the physical symmetries of the problem; time evolution is implemented through a Suzuki–Trotter decomposition, and dynamical spin-spin correlations are extracted with a Hadamard-test circuit. The full pipeline is then repeated under a noise model calibrated on real quantum hardware, identifying the number of Trotter steps that balances the approximation error against the error accumulated from noise, and testing its stability against noise-aware reoptimization and asymmetric readout error. The results show that an ansatz aligned with the symmetries of the Hamiltonian drastically reduces the number of parameters required with respect to a generic ansatz, an advantage that persists, at increasing cost, when moving from two to three spins — a design criterion that is a promising candidate for extension to larger molecular systems.

**Keywords:** quantum simulation; spin systems; Variational Quantum Eigensolver (VQE); Suzuki–Trotter decomposition; quantum noise; IBM Quantum hardware.